Topic: Undetermined coefficients for higher order equations

Question: Find the general solution to the second order differential equation.

$$y'' + 4y' + 3y = 9t$$

Answer choices:

A $y(t) = c_1 e^{4t} + c_2 e^{3t} + t - 1$

B $y(t) = c_1 e^{-t} + c_2 e^{-3t} + 3t - 4$

C $y(t) = c_1 t e^{t} + c_2 t^2 e^{3t} - 3t$

D $y(t) = c_1 + c_2 e^{-t} + 6$

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Solution: B

Before we find the general solution, we need to find the complementary solution and the particular solution.

To find the complementary solution $y_c(t)$, we solve the homogeneous form of the differential equation by factoring and finding the roots of the characteristic polynomial.

$$r^2 + 4r + 3 = 0$$

$$(r + 1)(r + 3) = 0$$

The roots are $r_1 = -1$ and $r_2 = -3$, each with multiplicity 1, so the complementary solution is

$$y_c(t) = c_1 e^{-t} + c_2 e^{-3t}$$

To find the particular solution $y_p(t)$, our guess and its derivatives will be

$$y_p(t) = At + B$$

$$y_p'(t) = A$$

$$y_p''(t) = 0$$

We plug these into the differential equation and get

$$0 + 4(A) + 3(At + B) = 9t$$

$$4A + 3At + 3B = 9t$$

$$(3A)t + (4A + 3B) = 9t$$

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Equating coefficients gives us the system of equations

$$3A = 9$$

$$4A + 3B = 0$$

When we solve this system of equations, we find $A = 3$ and $B = -4$, so the particular solution will be

$$y_p(t) = 3t - 4$$

To find the general solution, we'll sum the complementary and particular solutions.

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1 e^{-t} + c_2 e^{-3t} + 3t - 4$$

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Topic: Undetermined coefficients for higher order equations

Question: Given the complementary solution, find the general solution to the second order differential equation.

$$y'' - 2y' = 3te^{-t} + 2e^{-t}$$

$$y_c(t) = c_1 + c_2e^{2t}$$

Answer choices:

A $y(t) = c_1 + c_2e^{2t} + (2t + 3)e^{-t}$

B $y(t) = c_1 + c_2e^{2t} + (3t + 2)e^{-t}$

C $y(t) = c_1 + c_2e^{2t} + (t + 2)e^{-t}$

D $y(t) = c_1 + c_2t^2e^{2t} + 3e^{-t}$

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Solution: C

Since $g(t)$ is the product of a polynomial and an exponential,

$$g(t) = 3te^{-t} + 2e^{-t}$$

$$g(t) = (3t + 2)e^{-t}$$

we guess that the particular solution has the form

$$y_p(t) = (At + B)e^{-t}$$

$$y_p(t) = Ate^{-t} + Be^{-t}$$

The first and second derivatives of $y_p(t)$ are

$$y_p'(t) = A(e^{-t} - te^{-t}) - Be^{-t}$$

$$y_p'(t) = -Ate^{-t} + (A - B)e^{-t}$$

and

$$y_p''(t) = -A(e^{-t} - te^{-t}) - (A - B)e^{-t}$$

$$y_p''(t) = Ate^{-t} + (-2A + B)e^{-t}$$

We plug these into the differential equation and simplify.

$$Ate^{-t} + (-2A + B)e^{-t} - 2(-Ate^{-t} + (A - B)e^{-t}) = 3te^{-t} + 2e^{-t}$$

$$Ate^{-t} - 2Ae^{-t} + Be^{-t} + 2Ate^{-t} - 2Ae^{-t} + 2Be^{-t} = 3te^{-t} + 2e^{-t}$$

$$3Ate^{-t} + (-4A + 3B)e^{-t} = 3te^{-t} + 2e^{-t}$$

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Equating coefficients gives a system of equations.

$$3A = 3$$

$$-4A + 3B = 2$$

Solving the system gives $A = 1$ and $B = 2$, so the particular solution is

$$y_p(t) = te^{-t} + 2e^{-t}$$

$$y_p(t) = (t + 2)e^{-t}$$

To find the general solution, we'll sum the complementary and particular solutions.

$$y(t) = y_c(t) + y_p(t$$

$$y(t) = c_1 + c_2e^{2t} + (t + 2)e^{-t}$$

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Topic: Undetermined coefficients for higher order equations

Question: Find the best guess for the form for the particular solution of the fourth order differential equation.

$$y^{(4)} - 8y''' + 42y'' - 104y' + 169y = e^{2t}\sin(3t)$$

Answer choices:

A $y_p(t) = Ae^{2t}\cos(3t) + Be^{2t}\sin(3t)$

B $y_p(t) = Ae^{2t}\sin(3t)$

C $y_p(t) = Ate^{2t}\cos(3t) + Bte^{2t}\sin(3t)$

D $y_p(t) = At^2e^{2t}\cos(3t) + Bt^2e^{2t}\sin(3t)$

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Solution: D

The first step is to find the complementary solution. We need to factor the characteristic polynomial of the differential equation and find its roots.

$$r^4 - 8r^3 + 42r^2 - 104r + 169 = 0$$

$$(r^2 - 4r + 13)^2 = 0$$

The roots are $r_{1,2} = 2 \pm 3i$ with multiplicity 2, so the complementary solution is

$$y_c(t) = c_1 e^{2t} \cos(3t) + c_2 e^{2t} \sin(3t) + c_3 t e^{2t} \cos(3t) + c_4 t e^{2t} \sin(3t)$$

For the particular solution, since $g(t)$ is the product of a sine function and an exponential function, our guess should be

$$y_p(t) = Ae^{2t} \cos(3t) + Be^{2t} \sin(3t)$$

This guess overlaps with the first two terms of the complementary solution. If we multiply through by t,

$$y_p(t) = Ate^{2t} \cos(3t) + Bte^{2t} \sin(3t)$$

then the guess will overlap will the last two terms of the complementary solution. So we'll multiply through by t again to get the best version of our guess for the particular solution.

$$y_p(t) = At^2 e^{2t} \cos(3t) + Bt^2 e^{2t} \sin(3t)$$

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